MEMORANDUM

TO David / School Facilities CV Project **FROM** Rut Patel **DATE** September 3, 2026

SUBJECT A review-aware prototype for measuring school facilities from aerial imagery

Bottom line. On 67 date-matched field comparisons across nine audited schools, the frozen pipeline was correct on 45 (67.2%). Its average stated confidence was 70.1%. The original eight-school seeded sample was 40/59 (67.8%); Spring Lake was added openly as a solar-positive challenge and was 5/8. This is a small development audit of the submitted run, not a national accuracy estimate.

Result	N	Observed outcome	Why it matters
All scored fields	67	45 correct (67.2%)	Exact field-level score
Confidence 0.9	32	30 correct (93.8%)	Mostly matches its promise
Confidence 0.7	22	15 correct (68.2%)	Matches its promise closely
Confidence 0.4/0.1	13	0 correct	Correctly identifies highest risk
Review confidence <= 0.7	35	20/22 errors found	Leaves 30/32 automatic fields correct

APPROACH

1. Confirm the campus. I compared each supplied CCD point with one Nominatim candidate, opened the map links for all 25 schools, kept 24 CCD centers, and corrected one.

2. Use dated imagery. I downloaded NAIP through Microsoft Planetary Computer and saved the date, source item, crop, pixel scale, and coverage.

3. Show context and detail. One Gemini request receives the full campus, a centered close-up, and four overlapping sections.

4. Require structured evidence. The prompt separates rooftop solar from carports and detached portables from permanent buildings. Code converts solar image boxes to area using the saved pixel scale.

5. Lower confidence when evidence is weak. Missing edges, unclear ownership, poor visibility, stale imagery, or conflicting answers trigger review.

VALIDATION METHOD

A fixed seed selected eight schools by level (3 primary, 2 middle, 3 high). I saved predictions before labeling. Three helped improve the method; I later inspected the other five while comparing models, so I do not call them an untouched test.

I added Spring Lake after seeing rooftop solar so the audit contained one positive solar case. I made the final manual decisions and used ChatGPT/Claude only as second opinions for unfamiliar features. Unresolved fields were left blank.

Categories and counts require an exact match. Solar area would pass within 25% of a reliable manual measurement. Five date-mismatched or imprecise solar references remain visible but are not scored.

Cerra Vista's seven portable buildings are visible in the same 2022 NAIP image: five at the north edge and two near the center. Low contrast makes the gaps hard to see, but the label is scored. The model's zero count at 0.9 confidence is a real miss.

WHAT CONFIDENCE AND FLAGS SAY

The practical question is: **when the model is wrong, does it ask for help?** The 0.9, 0.7, 0.4, and 0.1 groups were compared with their actual success rates. They rank risk well, but Cerra and one track answer show that the high group is not perfect.

ECE is the weighted average absolute gap within confidence bands. It is 6.6 percentage points. The Brier score is 0.113 and penalizes confident mistakes. I report both because the overall 70.1% confidence versus 67.2% accuracy can hide a bad individual band.

Reviewing only 0.4/0.1 checks 13/67 fields and finds 13/22 errors. Reviewing <= 0.7 checks 35/67 and finds 20/22; the 32 fields left automatic are 30/32 correct. These are same-audit trade-offs, not deployment guarantees.

MAIN FAILURES

Fencing caused 13/22 errors. Thin lines, trees, shadows, incomplete Street View, and uncertain campus edges also make manual fence labels imperfect. Houses, hedges, and roads are boundaries, not fences.

Portable detection missed Cerra Vista's only positive case. Solar presence was 7/7 on date-matched labels, but only one was positive. Solar area is approximate and kept at low confidence. I did not complete the optional two-date measure because the Google dates were unknown. Ridgeview uses 2017 NAIP. Hosted runs had 503/429 errors, output variation, variable latency, and invalid responses.

HOW I WOULD SPEND 100 HOURS

Hours 1-20 - learn and define. Read work on solar mapping, building detection, facility counts, and confidence; compare imagery dates, resolution, licenses, and APIs; tighten the label guide; double-label a pilot to learn where people disagree.

Hours 21-40 - better campus boundaries. Test OSM/Overture buildings and land use, roads, open parcels, Microsoft Building Footprints, and district GIS layers. Review uncertain polygons in QGIS or a small web map.

Hours 41-65 - a real benchmark. Label 500-1,000 schools across geography, level, rurality, image age, and complexity. Include more solar, portables, pools, and fence types; use two labelers; keep separate development, confidence-adjustment, and untouched test schools.

Hours 66-90 - methods by attribute. Compare the VLM with building masks and specialized solar, portable, court, field, and track detectors (for example SAM 2, Grounding DINO, or YOLO after license checks). Treat fences separately with better aerial and legally usable street-level imagery.

Hours 91-100 - review rules and operations. Measure what each confidence level means on untouched labels, set the review cutoff from staff capacity and error cost, and add batching, caching, resumable jobs, data versions, latency, and cost monitoring.

SCALE, COST, AND API ACCESS

The final run used a median 8.76 seconds and about 8,125 input plus 432 output tokens per school. At the observed load, 130,000 schools cost about $348 with 3.1 Flash-Lite standard or $174 in batch. The completed 3.5 test gained only 1/58 correct fields, was slower, had slightly worse calibration, and implies about $5,404 standard. A 3.8 estimate is about $1,000 at the 3.1 token load, but repeated 503s prevent an accuracy claim.

A free-tier Gemini API key is sufficient for the 25-school assignment run. I used a personal Google AI Studio account because institutional Workspace access can be administrator-controlled. My observed 500 requests/day limit would take at least 260 days for 130,000 schools before retries, so free tier is not a scaling plan.

TIME AND AUTHORSHIP

Approximate time: 20 min understanding; 50 min approach and initial code; 30 min campus review; 60 min labeling the original eight; 90 min reliability checks, prompt changes, fixes, and review; 60 min model/final tests including the added solar case; 30 min memo and packaging. **Total: 5 h 40 min.**

I knowingly used the extra 40 minutes to learn unfamiliar aerial features, resolve labels, and investigate failures rather than rerun for a better-looking score. ChatGPT helped with initial design/code and visual second opinions; Claude with visual second opinions; Codex with code review, testing, prompt revisions, and packaging. I reviewed every change and remain responsible for definitions, campus decisions, labels, experiments, and reporting.

DATA, MODEL, SOFTWARE, AND SOURCES

CCD supplied starting coordinates; NAIP came from Microsoft Planetary Computer; Nominatim supplied cached candidates; Google Maps/Street View supported manual validation only; Gemini 3.1 Flash-Lite performed extraction. Python, pandas, rasterio, pyproj, Pillow, Pydantic, and pytest support the pipeline.

[1] planetarycomputer.microsoft.com/dataset/naip [2] operations.osmfoundation.org/policies/nominatim/ [3] docs.overturemaps.org/ [4] ai.google.dev/gemini-api/docs/get-started and /pricing

CALIBRATION TAKEAWAY

The 0.9 and 0.7 bands were close to their stated success rates, while the 0.4 band was overconfident (0/5 correct).
The ranking is useful for deciding what to review, but these values are not yet fully calibrated probabilities.